

**BEFORE YOU READ — ANSWER FROM MEMORY**

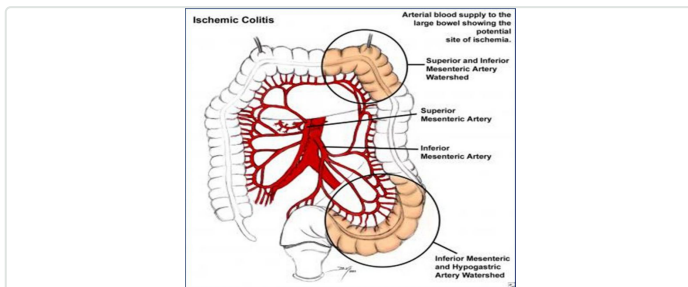
1. Which **two colonic regions** are the watershed zones most liable to ischemia?
2. Where is a biopsy taken to diagnose **celiac disease**, and what are the **three** histological features?
3. Which malabsorptive disease is the **only** one with a lymphatic transport defect?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

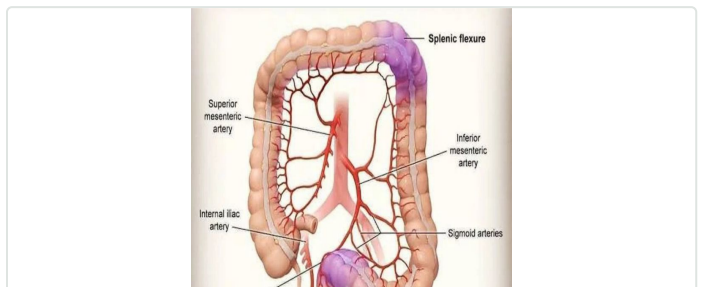
1 Ischemic bowel disease

Infarction of bowel from hypoperfusion. Mainly due to **impaired arterial supply**; uncommonly due to impaired venous drainage.

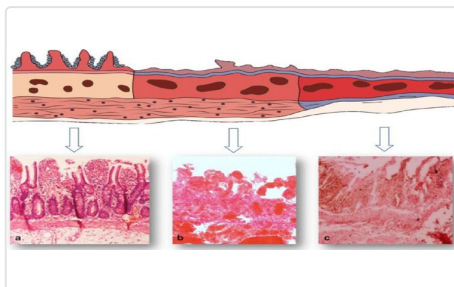
Mechanism	Causes
Non-obstructive arterial hypoperfusion	e.g. hypovolemic shock
Arterial obstruction	Severe atherosclerosis · aortic aneurysm · hypercoagulable states · oral contraceptive use · embolisation of cardiac vegetations or aortic atheroma · vasoconstrictive drugs · systemic vasculitides
Impaired venous drainage	Mesenteric venous thrombosis ; also invasive neoplasms, portal hypertension, trauma
Watershed zones	Most liable to ischemia: splenic flexure and sigmoid colon .
Morphology (depth)	(a) Mucosal infarction → (b) Mural : mucosa and submucosa → (c) Transmural : all three layers.



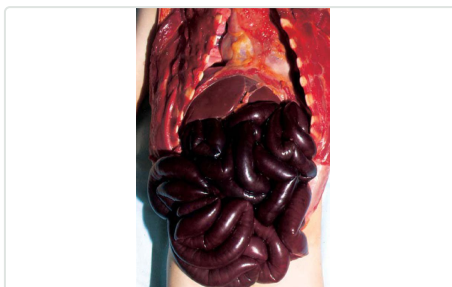
Arterial supply of the large bowel and its watershed zones



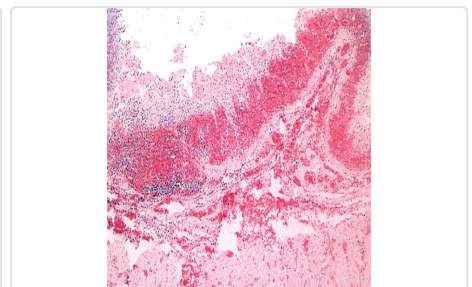
Splenic flexure and sigmoid: the watershed sites



Mucosal (a), mural (b), transmural (c) infarction



Gross: infarcted bowel



Microscopic: ischemic necrosis

2 Malabsorption syndrome

Inability to absorb nutrients, vitamins and minerals from the intestinal tract into the bloodstream.

General features	Diarrhea · steatorrhea · flatus · abdominal pain · weight loss.
Anemia	Iron, folate, vitamins B6 and B12.
Bleeding / purpura	Vitamin K.
Osteopenia, tetany	Calcium, magnesium, vitamin D.
Edema	Albumin.
Where absorption fails	Intraluminal digestion (salivary, gastric, pancreatic, small bowel, bile salts) · terminal (mucosal) digestion (brush border hydrolysis of complex carbohydrates, protein, fat) · transepithelial and lymphatic transport.

Defects in malabsorptive disease — the discriminator grid (slide; required diseases plus chronic pancreatitis)

Disease	Intraluminal digestion	Terminal digestion	Transepithelial transport	Lymphatic transport
Celiac disease		+	+	
Tropical sprue		+	+	
Cystic fibrosis	+			
Chronic pancreatitis	+			
Whipple disease				+
Abetalipoproteinemia			+	

Read the grid as: Whipple is the only lymphatic defect; CF and chronic pancreatitis hit intraluminal digestion only; abetalipoproteinemia is transepithelial only; celiac and tropical sprue hit terminal digestion and transepithelial transport.

DR JAWHER'S REQUIRED DISEASES — NOT COVERED ON THE SLIDES (from class notes; may come as cases)

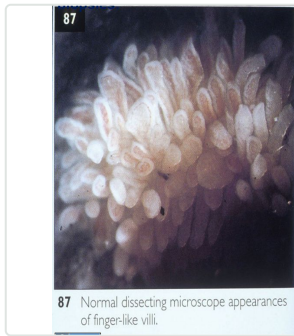
Tropical sprue	Diarrheal disease caused by infection (mostly E. coli, Klebsiella, Enterobacter).
Cystic fibrosis	Autosomal recessive ; malabsorption from lack of mucus clearance.
Whipple disease	<i>Tropheryma whipplei</i> (Gram positive); unlike tropical sprue, a systemic infection. Biopsy: macrophages in the lamina propria filled with bacteria.
Abetalipoproteinemia	Autosomal recessive ; β -lipoprotein is needed for lipid transport, so its absence causes lipid accumulation → steatorrhea.

3 Celiac disease — illness script

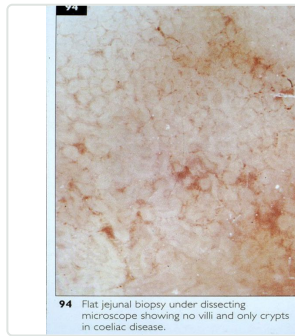
Also called non-tropical sprue or gluten-sensitive enteropathy: an **immune-mediated enteropathy** triggered by gluten-containing cereals (**wheat, rye, barley**) in genetically predisposed people.

Gluten → gliadin → Deamidated by tTG → HLA-DQ2 / DQ8 on APC → CD4+ T cells: cytokines → Tissue damage

WHO gets it	Genetically predisposed (HLA-DQ2 or HLA-DQ8). Infants: failure to thrive, diarrhea . Adults: diarrhea, flatulence, malabsorption, weight loss, fatigue.
WHY it happens	Gliadin is deamidated by tissue transglutaminase , presented on HLA-DQ2/DQ8 to CD4+ T cells (cytokines → damage). Enterocyte IL-15 activates CD8+ T cells (MIC-A / NKG2D) that kill enterocytes. B cells make anti-tTG, anti-deamidated gliadin and anti-endomysial antibodies — diagnostically useful.
WHAT you see	Biopsy of the second part of the duodenum or proximal jejunum : increased intraepithelial CD8+ T lymphocytes, crypt hyperplasia, villous atrophy .
Associations	Dermatitis herpetiformis · enteropathy-associated T-cell lymphoma (aggressive tumour of intraepithelial T cells) · increased frequency of small intestinal adenocarcinoma · splenic atrophy (unknown reason).
Treatment	Gluten-free diet.



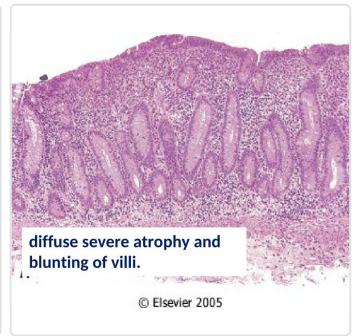
Normal villi (dissecting microscope)



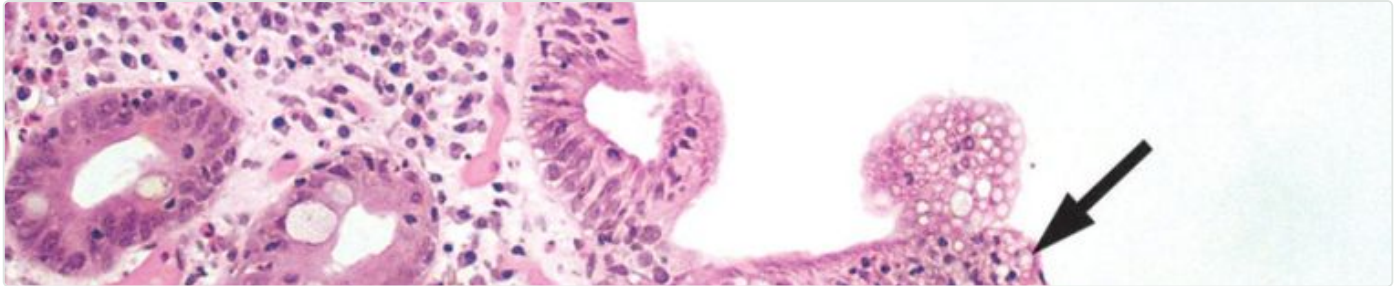
Flat mucosa: no villi, only crypts



Normal biopsy



Severe villous atrophy



Lymphocytic infiltrate (arrows: intraepithelial lymphocytes)

P Must-remember pearls

- **Watershed zones = splenic flexure and sigmoid colon.** Infarction ladder: mucosal → mural (mucosa + submucosa) → transmural.
- **Malabsorption grid:** Whipple = lymphatic only; CF and chronic pancreatitis = intraluminal only; abetalipoproteinemia = transepithelial only; celiac and tropical sprue = terminal digestion + transepithelial.
- **Celiac triad of histology:** intraepithelial CD8+ lymphocytes, crypt hyperplasia, villous atrophy — in the 2nd part of the duodenum or proximal jejunum.
- **Celiac = gluten (wheat, rye, barley) + HLA-DQ2/DQ8 + tTG;** diagnostic antibodies: anti-tTG, anti-deamidated gliadin, anti-endomysial; treatment is a gluten-free diet.
- **Celiac associations:** dermatitis herpetiformis, enteropathy-associated T-cell lymphoma, small intestinal adenocarcinoma, splenic atrophy.

Q Test yourself

1. A patient in hypovolemic shock develops bloody diarrhea. Which colonic regions are most liable to ischemia?
 - A. Splenic flexure and sigmoid colon
 - B. Cecum and ascending colon
 - C. Transverse colon and hepatic flexure
 - D. Rectum only
 - E. Descending colon only
2. A duodenal biopsy from a patient with diarrhea and weight loss shows increased intraepithelial CD8+ lymphocytes, crypt hyperplasia and villous atrophy. Which is the most useful serological marker?
 - A. Anti-mitochondrial antibody
 - B. Anti-smooth muscle antibody
 - C. Antinuclear antibody
 - D. Anti-tissue transglutaminase antibody
 - E. Anti-parietal cell antibody
3. A malabsorption patient has easy bruising and purpura. Which deficiency is responsible?
 - A. Vitamin B12
 - B. Calcium
 - C. Vitamin K
 - D. Albumin
 - E. Iron

4. Which malabsorptive disease shows a defect of lymphatic transport only?

- A.** Celiac disease
- B.** Cystic fibrosis
- C.** Abetalipoproteinemia
- D.** Tropical sprue
- E.** Whipple disease

5. Infarction involving the mucosa and submucosa but not the muscularis propria is best described as:

- A.** Mucosal infarction
- B.** Mural infarction
- C.** Transmural infarction
- D.** Serosal infarction
- E.** Watershed infarction

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 • A — The watershed zones, most liable to ischemia, are the splenic flexure and the sigmoid colon.

2 • D — Celiac disease: B cells produce anti-tTG, anti-deamidated gliadin and anti-endomysial antibodies, which are diagnostically useful.

3 • C — Bleeding/purpura reflects vitamin K deficiency. Anemia = iron, folate, B6, B12; osteopenia/tetany = Ca, Mg, vitamin D; edema = albumin.

4 • E — On the slide grid, Whipple disease is the only disease with a lymphatic transport defect.

5 • B — Morphology: mucosal (a), mural = mucosa and submucosa (b), transmural = all three layers (c).

SPACED REVIEW

☐ Day 1

☐ Day 3

☐ Day 7

☐ Day 21

☐ Pre-exam

Pathology GIT Block · MEDucation by Mattin Majid · summarised from SGL2, Dr Jawher Tahir Omer, Hawler Medical University
Study alongside the lecture, not instead of it.